

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech. Winter 2022-23 Examination

Semester: 8**Subject Code: 203105483****Subject Name: Data Science****Date: 07/10/2022****Time: 02:00 pm to 04:30 pm****Total Marks: 60****Instructions:**

1. All questions are compulsory.
2. Figures to the right indicate full marks.
3. Make suitable assumptions wherever necessary.
4. Start new question on new page.

Q.1 MCQ / Fill in the Blanks**(15)**

1. _____ is used to describe population growth over time or chemical reactions.
a.) Logistic distribution b.) Log normal distribution
c.) F distribution d.) Beta distribution
2. Which of the following approach should be used to ask Data Analysis question?
a.) Find only one solution for particular problem
b.) Find out the question which is to be answered
c.) Find out answer from dataset without asking question
d.) None of the mentioned
3. _____ describes the amount of time between events occurring at random moments.
a.) Chi-Square distribution b.) Exponential distribution
c.) Gamma distribution d.) Gumbel distribution
4. _____ is recorded as the rank order of scores.
a.) Nominal Data b.) Special Data
c.) Normal Data d.) Ordinal data
5. Which of the following is not type of the Secondary Data Sources?
a.) Internet Browsing Data b.) Satellite Data
c.) Interview Data d.) Sensor Data
6. Data science is the process of diverse set of data through?
a.) Organizing data b.) Processing data
c.) Analyzing data d.) All of the above
7. Which of the following is correct skills for a Data Scientist?
a) Probability & Statistics
b) Machine Learning / Deep Learning
c) Data Wrangling
d) All of the above
8. Data can be categorized into _____ groups.
a) 1 b) 2
c) 3 d) 4
9. A column is a _____ representation of data.
a) horizontal b) diagonal
c) vertical d) top
10. A _____ is a structured representation of data.
a.) database table b.) functions
c.) data preparation. d) data frame
11. What is Data Encoding?
12. Explain Visual Encoding.
13. Explain Mean.
14. Explain Median.
15. What is Variance?

- Q.2** Answer the following questions. (Attempt any three) **(15)**
- A) What is DATA? Explain with proper diagram.
 - B) What is Pandas library? Explain with example.
 - C) Explain Agile Framework in Data Science.
 - D) What is Data Visualization? Write down advantages of Data Visualization.
- Q.3** A) Explain Data Analysis Terminology& concepts with proper explanation. **(07)**
- B) Explain Linear Regression. **(08)**
- OR**
- B) Explain SVM in brief. **(08)**
- Q.4** A) Explain Naive Bayes in brief. **(07)**
- OR**
- A) Write down application and advantages of Naive Bayes **(07)**
- B) Explain Application development methods used in Data Science **(08)**